

MUHAMMAD DANISH

Full-Stack Developer | AI-Powered Applications | Claude API, LLM Integrations, RAG

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PROFESSIONAL SUMMARY

Full-stack developer building AI-powered web applications with React, Next.js, TypeScript, Node.js, and Python (FastAPI). Hands-on across the modern LLM stack: prompt engineering, system prompt design, tool use and function calling, retrieval-augmented generation (RAG), embeddings, streaming responses, and context window and token cost management across Anthropic Claude, Google Gemini, DeepSeek, and Perplexity APIs. Founder of Variorum, an AI-powered engineering knowledge platform combining multi-provider LLM orchestration, pgvector semantic search, and tree-sitter code analysis with secure GitHub App integrations. Core engineer on Recruitimate, an AI-native hiring platform spanning real-time interview intelligence and RAG-based resume screening, built with responsible AI guardrails aligned to emerging regulation. Delivers production features end to end, from API design through CI/CD and cloud deployment, and explains AI capabilities and limits clearly to non-technical stakeholders.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, Java, SQL, HTML, CSS

Frontend: React, Next.js (App Router, SSR), Tailwind CSS, Vite, React Router, Framer Motion, Recharts, responsive design, state management

Backend and APIs: Node.js, Express.js, FastAPI, SQLAlchemy, Alembic, Pydantic, Uvicorn (ASGI), REST API design, GraphQL, WebSockets, GitHub Apps, webhooks (HMAC verification), tree-sitter (AST parsing), rate limiting, structured error handling, JWT, OAuth, role-based access control, background jobs, microservices

AI / LLM: Anthropic Claude API, Google Gemini API, DeepSeek API, Perplexity API, Mistral AI, multi-provider LLM orchestration and fallback, prompt engineering, system prompt design, tool use and function calling, AI agents and multi-step workflows, retrieval-augmented generation (RAG), hybrid search (semantic + keyword), embeddings, streaming responses, context window and token cost management, Model Context Protocol (MCP), Claude Skills, guardrails and responsible AI, model selection and output evaluation

Databases and Vector Search: PostgreSQL, pgvector, full-text search (tsvector), Supabase, MongoDB, MySQL, schema design, query optimization

DevOps and Practices: Git, GitHub, Docker, docker-compose, CI/CD, Vercel, cloud deployment, pytest, Ruff, mypy, Jira, Agile / Scrum, code review, testing, performance optimization, observability

PROFESSIONAL EXPERIENCE

Junior Software Engineer

02/2026 - Present

Arken Systems

Core engineer on Recruitimate, an AI-native hiring platform: own features end to end across a Next.js and TypeScript frontend and a FastAPI backend on a multi-tenant PostgreSQL architecture.

Engineered multi-step interview intelligence workflows combining real-time audio over LiveKit (WebRTC), streaming transcription with AssemblyAI, and structured LLM scoring through tool use and function calling.

Built token-efficient prompt pipelines across Anthropic Claude and Google Gemini APIs, spanning system prompt design, context window management, streaming responses, and cost-aware model selection, with responsible AI guardrails aligned to the EU AI Act.

Implemented retrieval-augmented generation (RAG) for resume screening with an embedding-based retrieve-then-rerank pipeline on PostgreSQL with pgvector, cutting manual shortlisting effort by roughly half.

Frontend Engineer

02/2025 - 06/2026

RemoteFlow

Built responsive, reusable UI components in Next.js, React, and TypeScript for a scalable AI-powered job automation platform, following a server-first architecture.

Integrated frontend workflows with GraphQL and REST APIs, including streaming AI output and real-time status updates for long-running automation jobs.

Improved rendering and load performance through code splitting, caching, and component-level optimization, reducing key page load times by approximately 25%.

Delivered production-ready features in an agile team, with code reviews and shared quality standards that kept the codebase maintainable as the product grew.

Full Stack Developer

06/2025 - 09/2025

Baccalytics

Built a multi-tenant EdTech web platform for IB schools, teachers, and students using Next.js and Supabase (PostgreSQL, Auth, Storage).

Implemented role-based access control, school onboarding, and core LMS features for academic management and analytics.

Delivered a scalable, cloud-native platform from requirements through deployment, reducing manual academic administration for early school users.

Junior Software Developer

05/2024 - 12/2024

Alabtaal Developers

Developed a records management and performance reporting solution for a Madrassah alumni organization using Java Spring Boot.

Deployed services with Docker and managed delivery through Jira and GitHub in an agile workflow, improving deployment speed and cutting manual reporting effort.

SELECTED PROJECTS

Variorum | FastAPI, Next.js 15, React 19, TypeScript, PostgreSQL, pgvector, tree-sitter, Multi-LLM (Gemini, DeepSeek, Perplexity), RAG, GitHub Apps

Founder and lead engineer of Variorum, an AI-powered engineering knowledge platform that gives software teams a living memory of their codebase. It detects documentation drift, answers "why is the system built this way?" from commit and pull request history with cited sources, and flags untested, high-risk changes, then opens human-reviewed pull requests. Architected a provider-agnostic multi-LLM layer with automatic failover across Gemini, DeepSeek, and Perplexity, a hybrid RAG pipeline pairing pgvector embeddings with full-text search over code, docs, and history, deterministic tree-sitter AST indexing, and a secure GitHub App integration (RS256 JWT, OAuth, HMAC-verified webhooks) that re-indexes on every push.

FYP Idea Generator | Next.js, React, Mistral AI, CORE Research API, Supabase, Tailwind CSS

Research-grounded idea engine for final year students. Retrieves peer-reviewed literature through the CORE Research API and feeds it into an LLM pipeline that generates project ideas with problem statements, methodologies, build plans, and real references. Grounding generation in retrieved sources keeps citations verifiable, an applied RAG pattern.

Radiant Thought | React, TypeScript, Supabase, Gemini API, Tailwind CSS

AI journaling application with Supabase authentication and Gemini-powered entry titles and reflective insights covering themes, thought patterns, values alignment, and suggested actions. Structured prompts return consistent JSON for a clean, responsive reading experience.

EDUCATION

BS Computer Science (BSCS)

09/2022 - 06/2026

Air University, Aerospace and Aviation Campus Kamra | CGPA: 3.59 | Graduated June 2026

Relevant Coursework: Artificial Intelligence, Data Structures and Algorithms, Database Management Systems, Full Stack Development, Software Engineering, Data Science, Information Security, Computer Networks, Operating Systems

REFERENCES

Available upon request.